Problem 1. Let X and Y be two topological spaces and $f : X \to Y$.

(i) Prove that f is continuous if and only if $f^{-1}(C)$ is closed for all closed sets C in Y.

Proof. ($\Longrightarrow$) Suppose f is continuous. Let C be a closed set in Y. Our goal is to show that $f^{-1}(C)$ is closed in X. Since C is closed in Y, it follows that its complement C^c is open in Y. Since f is continuous, it preimage $f^{-1}(U)$ where $U = C^c$ is open in X. This tells us that $[f^{-1}(U)]^c$ is closed in X. Since $[f^{-1}(U)]^c = f^{-1}(U^c) = f^{-1}(C)$, it follows that $f^{-1}(C)$ is closed in X. ($\Longleftarrow$) Suppose that for all closed sets C in Y, $f^{-1}(C)$ is closed in X. Our goal is to show that f is continuous; that is, we want to show that for all open sets U in Y, $f^{-1}(U)$ is open in X. To this end, let U be an open set in Y. This tells us that $U^c = V$ is a closed set in Y. Then $f^{-1}(V)$ is closed in X by assumption. That is, $[f^{-1}(V)]^c$ is open in X. Since

$$[f^{-1}(V)]^c = f^{-1}(V^c) = f^{-1}(U),$$

it follows that $f^{-1}(U)$ is open in X which is our desired result. ■

(ii) Prove that f is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$ for all $A \subseteq X$.

Proof. ($\Longrightarrow$) Note that $\overline{f(A)}$ is a closed set in Y and so its preimage under f, $f^{-1}(\overline{f(A)})$, is closed in X from the previous part. Since $f(A) \subseteq \overline{f(A)}$, we have that $A \subseteq f^{-1}(\overline{f(A)})$. Since $f^{-1}(\overline{f(A)})$ is a closed set in X containing A and $\overline{A}$ is the smallest closed set in X containing A, we have that $\overline{A} \subseteq f^{-1}(\overline{f(A)})$, meaning that $f(\overline{A}) \subseteq \overline{f(A)}$.

($\Longleftarrow$) Let C be a closed set in Y. Our goal is to show that $f^{-1}(C)$ is closed in X; that is, we want to show that $\overline{f^{-1}(C)} = f^{-1}(C)$. Notice that $f^{-1}(C) \subseteq \overline{f^{-1}(C)}$ is immediate. So, it suffices to show that $\overline{f^{-1}(C)} \subseteq f^{-1}(C)$. Indeed, by setting $A = f^{-1}(C)$ which is a subset of X and using our assumption, we get that

$$\begin{aligned}
f(\overline{A}) \subseteq \overline{f(A)} &\Longrightarrow \overline{A} \subseteq f^{-1}(\overline{f(A)}) \\
&\Longrightarrow \overline{f^{-1}(C)} \subseteq f^{-1}\left[\overline{f(f^{-1}(C))}\right] \\
&\Longrightarrow \overline{f^{-1}(C)} \subseteq f^{-1}(\overline{C}) \\
&\Longrightarrow \overline{f^{-1}(C)} \subseteq f^{-1}(C) && (\overline{C} = C \text{ since } C \text{ is closed in } Y)
\end{aligned}$$

Since $\overline{f^{-1}(C)} \subseteq f^{-1}(C)$ and $f^{-1}(C) \subseteq \overline{f^{-1}(C)}$, we have $\overline{f^{-1}(C)} = f^{-1}(C)$ and so we conclude that $f^{-1}(C)$ is closed in X. Thus, f is continuous using part (i). ■

(iii) Prove that if, f is continuous and A is dense in X, then $f(A)$ is dense in $f(X)$.

Proof. Our goal is to show that $\overline{f(A)}^{f(X)} = f(X)$; that is, we want to show that $f(X) \subseteq \overline{f(A)}^{f(X)}$ and $\overline{f(A)}^{f(X)} \subseteq f(X)$. Starting with the first inclusion, we observe that since f is continuous and A is dense in X (i.e $\overline{A} = X$), it follows from part (ii) that (using a proposition proven in class)

$$\begin{aligned}
f(X) = f(\overline{A}) \subseteq \overline{f(A)}^{Y} &\Longrightarrow f(X) \subseteq \overline{f(A)}^{Y} \\
&\Longrightarrow f(X) \subseteq \overline{f(A)}^{Y} \cap f(X) \\
&\Longrightarrow f(X) \subseteq \overline{f(A)}^{f(X)}.
\end{aligned}$$

Now, the other inclusion follows immediately because, via the same proposition stated before, we have

$$\overline{f(A)}^{f(X)} = \overline{f(A)}^{Y} \cap f(X) \subseteq f(X).$$

Hence, it follows that $\overline{f(A)}^{f(X)} = f(X)$ and we conclude that $f(A)$ is dense in $f(X)$. ■

Problem 2 (Continuity and Lower Limit Topology). (i) Let $\mathcal{T}_L$ be the lower limit topology and $\mathcal{T}_E$ be the Euclidean topology on $\mathbb{R}$. Prove that $f : (\mathbb{R}, \mathcal{T}_L) \to (\mathbb{R}, \mathcal{T}_E)$ is continuous if and only if for every $x \in \mathbb{R}$ and $\varepsilon > 0$, there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for all $y \in [x, x+\delta)$.

Proof. ($\Longrightarrow$) Suppose f is continuous. Our goal is to show that for all $x \in \mathbb{R}$ and $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for all $y \in [x, x+\delta)$; that is, we need to show that for all $x \in \mathbb{R}$ and $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$[x, x+\delta) \subseteq f^{-1}(B_\varepsilon(f(x))).$$

Let $x \in \mathbb{R}$ and $\varepsilon > 0$ be given. Now, consider the open ball $B_\varepsilon(f(x))$ and notice that $x \in f^{-1}(B_\varepsilon(f(x)))$. Since $B_\varepsilon(f(x))$ is an open set in $(\mathbb{R}, \mathcal{T}_E)$ and f is continuous, we know that $f^{-1}(B_\varepsilon(f(x)))$ is open in $(\mathbb{R}, \mathcal{T}_L)$. Since $x \in f^{-1}(B_\varepsilon(f(x)))$, there exists a basis element U' containing x in $\mathcal{T}_L$ such that $x \in U' \subseteq f^{-1}(B_\varepsilon(f(x)))$. Since U' is a basis element, $U' = [a, b)$ for some $a, b \in \mathbb{R}$. Since $x \in U'$, define $\delta = \frac{1}{2}(b - x)$ (this is strictly positive because $x \in [a, b)$; that is, $a \leq x < b \Longrightarrow b - x > 0$). Thus, we see that $[x, x+\delta) \subseteq U' \subseteq f^{-1}(B_\varepsilon(f(x)))$ and so we conclude that

$$[x, x+\delta) \subseteq f^{-1}(B_\varepsilon(f(x))).$$

($\Longleftarrow$) Conversely, we will show that f is continuous. Suppose that for all $x \in \mathbb{R}$ and $\varepsilon > 0$, there exists a $\delta > 0$ such that $[x, x+\delta) \subseteq f^{-1}(B_\varepsilon(f(x)))$. To show that f is continuous, we need to show that for all open sets U in $(\mathbb{R}, \mathcal{T}_E)$, $f^{-1}(U)$ is open in $(\mathbb{R}, \mathcal{T}_L)$.

To this end, let U be an open set with respect to $\mathcal{T}_E$. Let $x \in f^{-1}(U)$ be arbitrary. Then $f(x) \in U$, by definition. Since U is open in $\mathcal{T}_E$, there exists an $\varepsilon > 0$ such that $B_\varepsilon(f(x)) \subseteq U$ with respect to the standard metric on $\mathbb{R}$. By assumption, since $x \in \mathbb{R}$ and $\varepsilon > 0$,

$$\text{there exists a } \delta > 0 \text{ such that } [x, x+\delta) \subseteq f^{-1}(B_\varepsilon(f(x))).$$

Since $[x, x+\delta) \in \mathcal{T}_L$ (that is, $[x, x+\delta)$ is open with respect to the Lower Limit Topology) contains x, it follows that

$$x \in [x, x+\delta) \subseteq f^{-1}(B_\varepsilon(f(x))) \subseteq f^{-1}(U) \Longrightarrow x \in [x, x+\delta) \subseteq f^{-1}(U).$$

Thus, this implies that $f^{-1}(U)$ is open in $\mathcal{T}_L$ and so f is continuous. ■

(ii) Is $f : (\mathbb{R}, \mathcal{T}_L) \to (\mathbb{R}, \mathcal{T}_E)$, $f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$ continuous?

Solution. We claim that f is continuous. We will show that for all $U \in \mathcal{T}_E$, the preimage under $f^{-1}(U)$ is open with respect to the lower limit topology. Indeed, let U be an open set in $\mathcal{T}_E$. Since the image under f only consists of either 1 or 0, we will consider four cases:

(1) Both 0 and 1 are not in U;

(2) $0 \in U$ but $1 \notin U$;

(3) $0 \notin U$ and $1 \in U$;

(4) $0 \in U$ and $1 \in U$.

Starting with (1), we notice that there is no $x \in \mathbb{R}$ such that $f(x) \in U$ and so $f^{-1}(U) = \emptyset$ which is clearly in $\mathcal{T}_L$.

Assuming (2), we notice that $f(x) = 0$ if and only if $x < 0$. Thus, the preimage of U under f is $f^{-1}(U) = (-\infty, 0)$ which can be written as a union of basis elements of $\mathcal{T}_L$ in the following way:

$$(-\infty, 0) = \bigcup_{n=1}^{\infty} [-n, 0)$$

which is clearly open since $[-n, 0)$ are open for all $n \in \mathbb{N}$ with respect to the Lower Limit Topology. Thus, $f^{-1}(U)$ is open.

Assuming (3), we notice $f(x) = 1$ if and only if $x \geq 0$. Thus, $f^{-1}(U) = [0, \infty)$ which is in $\mathcal{T}_L$. Thus, $f^{-1}(U)$ is open and so f is continuous.

Lastly, assume (4). We observe that since both 1 and 0 are in U, it follows that the preimage under f of U is just $\mathbb{R}$ itself which is contained in $\mathcal{T}_L$. Thus, $f^{-1}(U) = \mathbb{R}$ is open in $\mathcal{T}_L$. Thus, in all four cases, we get that $f^{-1}(U)$ is open in $\mathcal{T}_L$ and so we conclude that f is continuous. ■

(iii) Is $f : (\mathbb{R}, \mathcal{T}_E) \to (\mathbb{R}, \mathcal{T}_L)$, $f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$ continuous?

Solution. We claim that this function is not continuous. Take the open set $[1, 2)$ with respect to the Lower Limit Topology and observe that its pre-image under f will be

$$\begin{aligned} f^{-1}[1,2) &= \{x \in \mathbb{R} : f(x) \in [1,2)\} \\ &= \{x \in \mathbb{R} : 1 \leq f(x) < 2\} \\ &= \{x \in \mathbb{R} : x \geq 0\} \\ &= [0, +\infty) \end{aligned}$$

which is not open with respect to the Euclidean Topology since 0 is not an interior point of $[0, +\infty)$. ■

Problem 3 (Pull-back topology)**.** Let X be a set, $(Y, \mathcal{T})$ be a topological space, and $f : X \to Y$ be a function.

(i) Prove that $f^*\mathcal{T} = \{f^{-1}(U) : U \in \mathcal{T}\}$ is a topology on X. (This is called the pull-back topology on X induced by f.)

Proof. Our goal is to show the following conditions:

(i) $\emptyset, X \in f^*\mathcal{T}$;

(ii) For all $A_1, A_2 \in f^*\mathcal{T}$, $A_1 \cap A_2 \in f^*\mathcal{T}$;

(iii) If $\{A_\alpha\}_{\alpha \in \Omega}$ is an arbitrary collection in $f^*\mathcal{T}$, then $\bigcup_{\alpha \in \Omega} A_\alpha \in f^*\mathcal{T}$.

(i) Notice that $f^{-1}(\emptyset) = \emptyset$ and $f^{-1}(Y) = X$ where $\emptyset, Y \in \mathcal{T}$. Thus, it follows that $\emptyset, X \in f^*\mathcal{T}$.

(ii) Let $A_1, A_2 \in f^*\mathcal{T}$. Then by definition, we have $A_1 = f^{-1}(U_1)$ and $A_2 = f^{-1}(U_2)$ for some U_1 and U_2 in $\mathcal{T}$. Then

$$A_1 \cap A_2 = f^{-1}(U_1) \cap f^{-1}(U_2) = f^{-1}(U_1 \cap U_2).$$

Notice that since U_1 and U_2 are in $\mathcal{T}$ and $\mathcal{T}$ is a topology on Y, it follows that $U_1 \cap U_2 \in \mathcal{T}$. Thus, it follows that $A_1 \cap A_2 \in f^*\mathcal{T}$.

(iii) Let $\{A_\alpha\}_{\alpha \in \Omega}$ be a collection of sets in $f^*\mathcal{T}$. Then for all $\alpha \in \Omega$, $A_\alpha = f^{-1}(U_\alpha)$ for some $U_\alpha \in \mathcal{T}$. Then

$$\bigcup_{\alpha \in \Omega} A_\alpha = \bigcup_{\alpha \in \Omega} f^{-1}(U_\alpha) = f^{-1}\Big(\bigcup_{\alpha \in \Omega} U_\alpha\Big).$$

Since $\{U_\alpha\}_{\alpha \in \Omega} \subseteq \mathcal{T}$ and $\mathcal{T}$ is a topology, it follows that $\bigcup_{\alpha \in \Omega} U_\alpha \in \mathcal{T}$. This tells us that

$$\bigcup_{\alpha \in \Omega} A_\alpha \in f^*\mathcal{T}.$$

Thus, from (i) through (iii), we conclude that $f^*\mathcal{T}$ is, indeed, a topology on X. ■

(ii) Prove that $f : (X, f^*\mathcal{T}) \to (Y, \mathcal{T})$ is continuous.

Proof. Let U be an open set in $(Y, \mathcal{T})$ and let $x \in f^{-1}(U)$ and so, $f(x) \in U$. Since U is open in $(Y, \mathcal{T})$, there exists an open set W in $(Y, \mathcal{T})$ containing $f(x)$ such that $f(x) \in W \subseteq U$. But this tells us that $x \in f^{-1}(W) \subseteq f^{-1}(U)$ where $f^{-1}(W)$ is open in $(X, f^*\mathcal{T})$. Hence, it follows that $f^{-1}(U)$ is an open set in $(X, f^*\mathcal{T})$. Thus, f is continuous. ■

(iii) Prove that, if S is a topology on X such that $f : (X, S) \to (Y, \mathcal{T})$ is continuous, then $f^*\mathcal{T} \subseteq S$.

Proof. Let $A \in f^*\mathcal{T}$ be arbitrary. Then it follows by definition that there exists an open set U in $(Y, \mathcal{T})$ such that $A = f^{-1}(U)$. Since $f : (X, S) \to (Y, \mathcal{T})$ is continuous and U is an open set in $(Y, \mathcal{T})$, $f^{-1}(U)$ is open in (X, S), but this tells us that A is open with respect to S in X; that is, $A \in S$. Hence, $f^*\mathcal{T} \subseteq S$. ■

Remark. In the next problem, whenever I mention an open set in $\mathbb{R}^n$ where $n \in \mathbb{N}$, I am assuming with respect to the Euclidean Topology on $\mathbb{R}^n$.

Problem 4 (Open and Closed Sets Via Continuous Functions). (i) Prove that $\{x \in \mathbb{R} : \cos x > \sin x\}$ is open in $\mathbb{R}$.

Proof. Define $f(x) = \cos x - \sin x$ where $f : \mathbb{R} \to \mathbb{R}$. Since $\sin x$ and $\cos x$ are continuous on $\mathbb{R}$ and that the difference of continuous functions is continuous, it follows that $f(x)$ is continuous. Consider the open set $(0, +\infty)$ in $\mathbb{R}$. We see that

$$\begin{aligned} f^{-1}(0, +\infty) &= \{x \in \mathbb{R} : f(x) \in (0, +\infty)\} \\ &= \{x \in \mathbb{R} : f(x) > 0\} \\ &= \{x \in \mathbb{R} : \cos x - \sin x > 0\} \\ &= \{x \in \mathbb{R} : \cos x > \sin x\}. \end{aligned}$$

Since f is continuous and $(0, +\infty)$ is open in $\mathbb{R}$, we know that $f^{-1}((0, +\infty))$ is open in $\mathbb{R}$. That is, $x \in \mathbb{R} : \cos x > \sin x$ is open in $\mathbb{R}$. ■

(ii) Show that $\{(x, y) \in \mathbb{R}^2 : x > y,\ x^2 + y^2 < 1\}$ is open in $\mathbb{R}^2$.

Proof. First, note that the set above can be written as an intersection of the following two sets

(a) $\{(x, y) \in \mathbb{R}^2 : x > y\}$,

(b) $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$.

Thus, our goal is to show that the sets in (a) and (b) are open so that we can conclude that their intersection (because a finite intersection of open sets is open) is open.

(a) Define $f(x, y) = x - y$ where $f : \mathbb{R}^2 \to \mathbb{R}$. Since x and y are polynomials in $\mathbb{R}$ which are continuous, it follows that their difference $f(x, y)$ is also continuous. Consider the open set $(0, +\infty)$ in $\mathbb{R}$. Then it follows that

$$\begin{aligned} f^{-1}(0, +\infty) &= \{(x, y) \in \mathbb{R}^2 : f(x, y) \in (0, +\infty)\} \\ &= \{(x, y) \in \mathbb{R}^2 : f(x, y) > 0\} \\ &= \{(x, y) \in \mathbb{R}^2 : x - y > 0\} \\ &= \{(x, y) \in \mathbb{R}^2 : x > y\}. \end{aligned}$$

Since f is continuous and $(0, +\infty)$ is open in $\mathbb{R}$, it follows that $f^{-1}(0, +\infty)$ is open in $\mathbb{R}^2$. That is, $\{(x, y) \in \mathbb{R}^2 : x > y\}$ is open in $\mathbb{R}^2$.

(b) Define $f(x, y) = x^2 + y^2$ where $f : \mathbb{R}^2 \to \mathbb{R}$. Notice that f is continuous because x^2 and y^2 are polynomials in $\mathbb{R}$ which are continuous and that the sum of these two continuous functions is continuous. Thus, f is continuous on $\mathbb{R}^2$. Now, consider the open set $(-\infty, 1)$ in $\mathbb{R}$. Since f is continuous, we know that its preimage $f^{-1}(-\infty, 1)$ is open in $\mathbb{R}^2$. Then

$$\begin{aligned} f^{-1}(-\infty, 1) &= \{(x, y) \in \mathbb{R}^2 : f(x, y) \in (-\infty, 1)\} \\ &= \{(x, y) \in \mathbb{R}^2 : f(x, y) < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}. \end{aligned}$$

Hence, $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$ is open in $\mathbb{R}^2$.

Now, the intersection of the sets in (a) and (b) are open and so the original set $\{(x, y) \in \mathbb{R}^2 : x > y, x^2 + y^2 < 1\}$ are open in $\mathbb{R}^2$. ∎

(iii) Prove that $\{(x, y, z) \in \mathbb{R}^3 : 4 \leq x^2 + y^2 + z^2 \leq 6\}$ is closed in $\mathbb{R}^3$.

Proof. Define $f : \mathbb{R}^3 \to \mathbb{R}$ by $f(x, y, z) = x^2 + y^2 + z^2$. Since x^2, y^2, and z^2 are polynomials in $\mathbb{R}$ which are continuous, it follows from the fact that a sum of continuous functions that f is continuous. Consider the closed set $[4, 6]$ in $\mathbb{R}$. Since f is continuous, its preimage $f^{-1}([4, 6])$ is closed in $\mathbb{R}^3$. That is,

$$\begin{aligned} f^{-1}([4, 6]) &= \{(x, y, z) \in \mathbb{R}^3 : f(x, y, z) \in [4, 6]\} \\ &= \{(x, y, z) \in \mathbb{R}^3 : 4 \leq f(x, y, z) \leq 6\} \\ &= \{(x, y, z) \in \mathbb{R}^3 : 4 \leq x^2 + y^2 + z^2 \leq 6\}. \end{aligned}$$

Hence, it follows that $\{(x, y, z) \in \mathbb{R}^3 : 4 \leq x^2 + y^2 + z^2 \leq 6\}$ is closed in $\mathbb{R}^3$. ∎

Problem 5 (Pasting Lemma For A Finite Cover). Assume the hypothesis of the pasting lemma. Show that if A_i is closed for all $i \in I$ and I is finite, then f is continuous.

Proof. Let U be closed in Y and let $\{A_i\}_{i \in I}$ be a finite cover of X; that is,

$$\bigcup_{i=1}^{n} A_i = X.$$

Our goal is to show that $f^{-1}(U)$ is closed in X. Notice that

$$\begin{aligned} f^{-1}(U) = f^{-1}(U) \cap X = f^{-1}(U) \cap \Big(\bigcup_{i=1}^{n} A_i\Big) \\ = \bigcup_{i=1}^{n} [f^{-1}(U) \cap A_i]. \end{aligned}$$

Note that since $f(x) = f_i(x)$ for all $x \in A_i$, we get that for all $1 \leq i \leq n$

$$\begin{aligned} f_i^{-1}(U) &= \{x \in A_i : f_i(x) \in U\} \\ &= \{x \in A_i : f(x) \in U\} \\ &= \{x \in A_i : x \in f^{-1}(U)\} \\ &= A_i \cap f^{-1}(U). \end{aligned}$$

Then we get that

$$f^{-1}(U) = \bigcup_{i=1}^{n} f_i^{-1}(U).$$

Since each f_i is continuous on each A_i and U is a closed set in Y, it follows that $f_i^{-1}(U)$ is closed in A_i. Since each A_i is closed in X, it follows that $f_i^{-1}(U)$ is closed in X for all $1 \leq i \leq n$. Since a finite union of closed sets is closed, it follows that

$$f^{-1}(U) = \bigcup_{i=1}^{n} f_i^{-1}(U)$$

is a closed set in X and so we conclude that f is continuous. ■

Problem 6. (i) Let X be a topological space and A is a subspace of X. Consider the inclusion $i : A \to X$, $i(x) = x$. Then

(a) A is open in X if and only if the inclusion i is open.

Proof. Let $\mathcal{T}$ be a topology on X and let $\mathcal{T}|_A$ be the subspace topology endowed from X.

($\Longrightarrow$) Suppose A is open. Our goal is to show that the inclusion map i is open; that is, $i(U) \in \mathcal{T}$ for all $U \in \mathcal{T}|_A$. Let $U \in \mathcal{T}|_A$ be arbitrary (that is, U is open in A). We want to show that $i(U)$ is open in X. Since U is open in A, there exists an open set V in X such that $U = V \cap A$. Since A and V are open in X, it follows that U must be open because a finite intersection of open sets is open. Thus, $i(U) = U$ implies that $i(U)$ must be open in X.

($\Longleftarrow$) Suppose that i is open. Then i is open if and only if for all $U \in \mathcal{T}|_A$, $i(U) \in \mathcal{T}$. In particular, $A \in \mathcal{T}|_A$, and so $i(A) \in \mathcal{T}$. But note that $i(A) = A$ and so $A \in \mathcal{T}$. Thus, it follows that A is open in X. ■

(b) A is closed in X if and only if the inclusion i is closed.

Proof. ($\Longrightarrow$) Suppose A is closed in X. Let U be a closed set in A. Then there exists a closed set V in X such that $U = V \cap A$. However, A and V are both closed in X which implies that $U = V \cap A$ is closed because the finite intersection of closed sets is closed. Thus, $i(U) = U$ is closed in X which is our desired result.

($\Longleftarrow$) Suppose that i is closed. That is, for all closed sets V in A, $i(V)$ is closed in X. Since A itself is closed in its subspace topology, it follows that $i(A)$ is closed in X. But by definition $i(A) = A$ and so A is closed in X.

■

(ii) Let $n, k \in \mathbb{N}$. Prove that the function

$$f : \mathbb{R}^n \to \mathbb{R}^{n+k} \quad f(x_1, x_2, \dots, x_n) = (x_1, x_2, \dots, x_n, 0, 0, \dots, 0)$$

is not open.

Proof. Equip $\mathbb{R}^n$ and $\mathbb{R}^{n+k}$ with the Euclidean Topology. Since $\mathbb{R}^n$ and $\mathbb{R}^{n+k}$ are metrizable topological spaces, we can equip both of them with the Euclidean norm $\|\cdot\|$. Define the open ball $U = B_\varepsilon(0)$ centered at 0 with an arbitrary radius $\varepsilon > 0$ in $\mathbb{R}^n$ which is clearly open with respect to the Euclidean Topology on $\mathbb{R}^n$. The image of this ball under f is

$$f(U) = \{(x_1, x_2, \dots, x_n, 0, 0, \dots, 0) \in \mathbb{R}^{n+k} : (x_1, x_2, \dots, x_n) \in B_\varepsilon(0)\}.$$

We claim that $0 \in f(U)$ is not an interior point of $f(U)$, implying that $f(U)$ is not open in $\mathbb{R}^{n+k}$. Let $r > 0$, and consider the ball $B_r(0) \subseteq \mathbb{R}^{n+k}$. Notice that this ball contains points that have at least one i-th component where $i > n$ that is non-zero. In particular, take the following point

$$y = (0, \dots, 0, \dots, 0, r/2, 0, \dots, 0) \in B_r(0).$$

We see that this point is contained in $B_r(0)$ since

$$\|y - 0\| = \Big(\sum_{i=1}^{n+k} |y_i - 0|^2\Big)^{1/2} = \Big|\frac{r}{2} - 0\Big| = \frac{r}{2} < r$$

but not contained in $f(U)$ because it contains some k-th coordinate that is non-zero where $k > n$ (namely, from the above $y_k = \frac{r}{2}$). Hence, there does not exists an open ball centered at 0 that is contained in $f(U)$. Therefore, $f(U)$ is not open in $\mathbb{R}^{n+k}$ and so f is not an open map. ■